How to Write a Lab Report

Structure, Format & Examples

A Thesis Guide

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2026

Abstract

A laboratory report is the standard way scientists communicate what they did, what they found, and what it means. Unlike an essay, it follows a fixed structure — the IMRaD structure of Introduction, Methods, Results and Discussion — so that any reader can find the information they need in seconds and can judge whether the work is sound. This guide explains that structure section by section, from the title and the abstract to the references and the appendices, and shows how to write each part in clear, precise scientific language.

The guide covers the practical skills that make a lab report convincing: writing a title that states the finding, an abstract that summarises the whole experiment in a few sentences, an introduction that states the aim and the theory, a methods section detailed enough for another student to repeat the experiment, results presented through well-designed tables and graphs, and a discussion that interprets the findings honestly — including a proper analysis of errors and uncertainties. Later chapters cover scientific style, tense and units, referencing, appendices, and the most common mistakes students make.

The final worked example presents a complete fictional experiment report, so that a student can see the whole structure working together before writing their own. The advice applies to lab reports in physics, chemistry, biology and the applied sciences at undergraduate level, and it can be adapted to the format required by any department or teacher.

Keywords: lab report, IMRaD, scientific writing, methods, results, discussion, error analysis, referencing.

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How to use this guide: read Chapters 1 and 2 before your first lab report, to understand the purpose and the structure. Then work through Chapters 3 to 8 in order as you write each section of your report. Use Chapters 9 to 11 while you revise, Chapter 12 as a model to follow, and Chapter 13 as a final check before submission.

Chapter 1: Purpose of a Lab Report

A lab report is a record of an experiment, written so that another person can understand what was done, what was found, and what it means — without having been in the laboratory. That is its whole purpose, and every rule in this guide follows from it. The reader may be a teacher marking the work, a fellow student repeating the experiment, or a researcher years later checking the result. The report must therefore be complete, accurate and honest: complete enough to repeat, accurate enough to trust, and honest enough to show where the evidence is weak.

A lab report is not a diary of what happened in the lab, and it is not a story. It does not describe how you felt about the experiment, or list every small thing you did in the order you did it. It selects and organises the information the reader needs: the aim, the theory behind it, the method, the results, and the interpretation. Anything that does not help the reader judge the experiment does not belong in the report. This discipline — writing only what the reader needs, in the order the reader expects — is the first skill of scientific writing, and it is the skill this guide teaches.

1.1 Who reads a lab report, and what do they look for?

At undergraduate level, the main reader is your teacher or demonstrator, who will mark the report against a set of criteria: Is the aim clear? Is the method described well enough to repeat? Are the results presented clearly and correctly? Does the discussion interpret the results, including their errors? Is the conclusion supported by the evidence? A good report makes the marker's job easy, because every criterion is met in the section where the marker expects to find it. A report that hides the results in the discussion, or the method in the introduction, forces the marker to search — and loses marks for it.

1.2 Honesty is the foundation

Science depends on trust, and a lab report is a promise that the work is reported as it happened. Record your measurements as you take them, in a notebook, with the date and the conditions. Never change a reading to make it fit the theory, never leave out a result because it looks wrong, and never report a measurement you did not take. An unexpected result is not a failure; it is information, and the discussion chapter is where you explain it. Students who invent or tidy their data may pass one report, but they learn nothing — and in real research, dishonesty ends careers. The habit of honest recording starts with the first lab report you write.

Worked Example: What a lab report is — and is not

It IS: a clear, organised account of one experiment, with an aim, a method, results and an interpretation, written so that the experiment could be repeated. It is NOT: a diary ('First I did this, then I did that'), a textbook chapter (long explanations of basic theory), or a story with suspense ('The results were surprising — read on to find out'). The reader should know the aim in the first paragraph and the main result in the abstract.

[ ] I know who will read my report and what they will mark.

[ ] I recorded my measurements honestly, as I took them, in a notebook.

[ ] I can state the aim of the experiment in one sentence.

[ ] I have all my raw data, with dates and conditions, before I start writing.

Chapter 2: The IMRaD Structure at a Glance

Almost every scientific report in the world follows the same structure, called IMRaD: Introduction, Methods, Results and Discussion. The structure is not a tradition for its own sake; it is a tool. A reader who has read one IMRaD report can read any other, because the information is always in the same place. The introduction says why the experiment was done. The methods say how it was done. The results say what was found. The discussion says what it means. Around this core sit the title, the abstract, the conclusion, the references and the appendices.

The order of the sections is the order of the logic, not the order of the work. In the laboratory you may have done the method first, but in the report the introduction comes first, because the reader needs to know why the work was done before they can judge how it was done. Similarly, the abstract is written last, because only when the report is finished do you know what it contains. Learn the structure once, and

every lab report you write for the rest of your studies will follow the same pattern.

Section	Question it answers	Written when
Title	What was studied?	Last
Abstract	What was done, found and concluded?	Last
Introduction	Why was this experiment done?	First, after planning
Methods	How was it done?	During / just after the lab
Results	What was found?	After analysis
Discussion	What does it mean?	After the results
Conclusion	What is the answer?	Last
References	Where did the ideas come from?	As you write
Appendices	What extra detail is needed?	Last

2.1 How long should each section be?

There is no fixed length for a lab report — a short undergraduate experiment may need only a few pages, while a final-year project may need twenty or more. But the proportions are fairly stable: the introduction is usually the longest text section, the methods are detailed but concise, the results are mostly tables and figures with short text, and the discussion is about as long as the introduction. The abstract is always short — usually 150 to 250 words. If one section is much longer than the others, ask yourself whether it is doing another section's job: a very long introduction may be repeating textbook theory, and a very long results section may be interpreting instead of presenting.

2.2 Write in order, revise in order

Write the sections in the order you can write them: methods first, while the experiment is fresh in your mind; then results, while the analysis is in front of you; then the discussion; then the introduction, now that you know exactly what the report claims; and finally the abstract, the title and the conclusion. This order feels backwards, but it works, because each section depends on the one before it. Many students write the introduction first and then discover that their results do not match

what they promised — and have to rewrite it. Writing the introduction last avoids that wasted effort.

[ ] I know the IMRaD order and what each section must do.

[ ] I will write the methods first, while the experiment is fresh.

[ ] I will write the abstract and the title last.

[ ] No section is doing another section's job.

Chapter 3: Title and Abstract

The title and the abstract are the smallest parts of a lab report and the most-read. A teacher may read dozens of reports, and the title and abstract are what tell them, in seconds, what the experiment was and what it found. Both are written last, when the rest of the report is finished, and both must be accurate: they are promises about the contents, and a reader who finds a different experiment inside the report will not trust the rest of it.

3.1 Writing a good title

A good title names the variable studied and the result or the method, in as few words as possible. It is specific, not general: 'The Effect of Temperature on the Rate of Reaction between Hydrochloric Acid and Calcium Carbonate' tells the reader exactly what was studied, while 'Chemistry Experiment 3' tells them nothing. Avoid titles that are questions, titles that are jokes, and titles that promise more than the experiment delivered. If the experiment tested one factor, the title should name that factor. A useful test: could a student find your report in a library by searching for the words in your title? If not, the title is too vague.

3.2 Writing the abstract

The abstract is a summary of the whole report in 150 to 250 words, usually in a single paragraph. It has four parts, in order: the aim (one sentence); the method (one or two sentences — what was done, with the key conditions); the main results (two or three sentences, with the key numbers); and the conclusion (one sentence — what the results mean). Use the past tense, avoid citations, and do not include material that is not in the report. A reader who reads only the abstract should know what you did, what you found, and what it means.

Worked Example: A strong abstract (fictional experiment)

'This experiment investigated how temperature affects the rate of reaction between hydrochloric acid and calcium carbonate. Five temperatures between 20 and 60 degrees Celsius were tested, and the time taken to collect 50 cubic centimetres of carbon dioxide was measured for each. The reaction time fell from 96 seconds at 20 degrees to 24 seconds at 60 degrees, and the rate approximately doubled for each 10-degree rise. The results support the theory that higher temperatures increase the rate of reaction by increasing the frequency and energy of collisions between particles.' In five sentences this abstract gives the aim, the method, the key numbers and the conclusion. That is the

standard to aim for.

[ ] The title names the variable studied and the method or result.

[ ] The abstract is 150-250 words, in a single paragraph.

[ ] The abstract gives the aim, the method, the main results and the conclusion.

[ ] The abstract contains the key numbers, not vague words like 'increased a lot'.

[ ] Nothing in the abstract is missing from the main report.

Chapter 4: Introduction and Theory

The introduction has two jobs: to explain why the experiment was done, and to give the reader the theory they need to understand it. It usually opens with the background — the scientific idea behind the experiment, stated briefly — then narrows to the specific question the experiment will answer, and ends with a clear statement of the aim and, where appropriate, a prediction (hypothesis) of what is expected. A reader who finishes the introduction should know exactly what the experiment will test and why it matters.

Keep the theory short. The introduction is not a textbook chapter, and it should not explain everything that is known about the topic. Include only the theory the reader needs to understand the method and the results: the key equation, the key relationship, or the key concept. If the experiment tests whether the rate of reaction increases with temperature, the reader needs the collision idea in a few sentences — not a full chapter on chemical kinetics. Long introductions that repeat textbook material are the most common weakness in student lab reports.

4.1 Stating the aim and the hypothesis

State the aim in one clear sentence, usually at the end of the introduction: 'The aim of this experiment was to investigate how temperature affects the rate of reaction between hydrochloric acid and calcium carbonate.' A hypothesis is a prediction based on theory, stated before the experiment: 'It was predicted that the rate of reaction would increase with temperature, because higher temperatures increase the energy of the particles.' Not every experiment needs a formal hypothesis — some are exploratory — but where the theory makes a clear prediction, state it. The discussion chapter will return to it and judge whether it was supported.

4.2 Referencing the background

If the introduction uses ideas from a textbook or another source, reference them. Even a short lab report must show where its ideas come from, and a claim without a reference — 'It is known that...' — is a claim the reader cannot check. Use the referencing style your department requires, and keep the reference list short: two or three sources are usually enough for an undergraduate lab report. Chapter 9 explains referencing in full.

[ ] The introduction explains why the experiment was done.

[ ] Only the theory needed to understand the experiment is included.

[ ] The aim is stated in one clear sentence.

[ ] A hypothesis is stated where the theory makes a clear prediction.

[ ] Ideas taken from other sources are referenced.

Chapter 5: Materials and Methods — Writing for Reproducibility

The methods section must allow another student, with the same equipment, to repeat your experiment and obtain similar results. That is its only test, and it is a demanding one. It must list the materials and apparatus, describe the procedure step by step, state the quantities and conditions, and explain how the measurements were taken and how many times they were repeated. A methods section that a reader cannot follow is a methods section that has failed, however good the experiment was.

Write the methods in the past tense and in paragraph form, not as a numbered list of instructions copied from the lab manual. The lab manual tells you what to do; the report tells the reader what you actually did, including any changes you made. If you changed a quantity, used different equipment, or repeated a measurement because the first reading looked wrong, say so. Examiners notice the difference between a report and a copied manual, and they mark accordingly.

5.1 Materials and apparatus

List the chemicals and their concentrations, the equipment and its sizes, and the measuring instruments with their precision. Give the numbers: '50 cm3 of 1.0 mol/dm3 hydrochloric acid', not 'some acid'. Name the instrument and its smallest scale division — for example, a thermometer reading to 0.5 degrees, or a balance reading to 0.01 g — because the precision of the instrument limits the precision of the result, and the discussion chapter will need it for the error analysis. A labelled diagram of the apparatus is often worth more than a paragraph of description, and it should be included where the set-up is not obvious.

5.2 Procedure

Describe the procedure in the order it happened, in enough detail to repeat, and no more. State the quantities used, the conditions (temperature, time, order of mixing), and the measurements taken at each stage. State how many repeats were made and why. Describe the control of variables: which factors were kept constant, and how. If the experiment had a control — a run with no change, for comparison — describe it. Do not include results in the methods section, and do not explain why a step was taken; the explanation belongs in the discussion.

5.3 Safety and risk

Many experiments involve hazards: hot liquids, corrosive chemicals, glassware under pressure. State the main hazards and the precautions taken — eye protection, gloves, working in a fume cupboard, keeping flammable materials away from flames. This is not decoration; it shows that the experiment was planned responsibly, and in

some departments it is a required part of the report. Keep it short: one paragraph, or a few lines, naming the hazard and the precaution for each.

Worked Example: A reproducible procedure (fictional experiment)

Weak: 'Acid was added to the calcium carbonate and the gas was collected.' The reader cannot repeat this — how much acid, what concentration, how was the gas collected, for how long? Strong: '50 cm3 of 1.0 mol/dm3 hydrochloric acid was placed in a conical flask in a water bath at 20 degrees Celsius. 5.0 g of calcium carbonate chips were added, the flask was stoppered, and the carbon dioxide produced was collected in a gas syringe. The volume was recorded every 10 seconds until 50 cm3 had been collected. The procedure was repeated at 30, 40, 50 and 60 degrees Celsius, with three repeats at each temperature.' The strong version can be repeated. That is the test.

[ ] All quantities, concentrations and conditions are stated with numbers.

[ ] Instruments are named, with their precision.

[ ] The procedure is described step by step, in the order it happened.

[ ] Repeats, controls and variables kept constant are all stated.

[ ] Any changes from the lab manual are reported honestly.

[ ] Safety hazards and precautions are stated briefly.

Chapter 6: Results: Tables, Graphs and Figures

The results section presents what you found — nothing more. It does not interpret the findings, compare them with theory, or explain why they happened. That is the discussion chapter's job. The results section's job is to give the reader a clear, accurate picture of the data, in tables and figures, with just enough text to guide the reader through them. The most common fault in student results sections is mixing results with interpretation, so that the reader cannot tell where the data end and the author's opinion begins.

Present the data once, in the best form for the reader. Use a table for precise numbers the reader may need to check, and a graph for a pattern or a relationship the reader should see at a glance. Never present the same data in both a table and a graph — choose the form that serves the reader best. The raw data belong in an appendix, not in the results section; the results section shows the processed data — the means, the rates, the values the experiment was designed to produce.

6.1 Tables

A good table is simple: a clear title above it, column headings with units, and numbers given to the right number of significant figures — the precision the instrument allows, not the ten digits the calculator produced. Show the mean of repeated measurements, and where appropriate the range or the uncertainty. Number the tables (Table 1, Table 2) and refer to each one in the text: 'Table 1 shows that the reaction time fell as the temperature rose.' A table the text never mentions is a table the reader will skip.

Temperature (C)	Time 1 (s)	Time 2 (s)	Time 3 (s)	Mean time (s)
20	98	95	96	96
30	68	66	67	67
40	47	45	46	46
50	33	32	34	33
60	25	24	23	24

Table 1 (above) is a fictional example from the temperature experiment described in this guide. Note what it does: the title would normally sit above the table; every column has a heading with its unit; the three repeats are shown, so the reader can see how consistent they are; and the mean — the value the discussion will use — is in the final column. In the text, the writer would draw attention to the pattern: the mean time fell from 96 seconds at 20 degrees to 24 seconds at 60 degrees.

6.2 Graphs

A good graph follows the same rules. It has a title, labelled axes with units, and points plotted clearly, with a line of best fit where the relationship is a straight line or a smooth curve. The independent variable — the one you changed, such as temperature — goes on the horizontal axis, and the dependent variable — the one you measured, such as reaction time — goes on the vertical axis. Choose a scale that fills the graph and makes the pattern visible; a graph squeezed into one corner tells the reader nothing. If you draw a line of best fit, say whether it is a straight line or a curve, and do not join the points dot-to-dot unless the data are genuinely connected in that way. Every figure is numbered (Figure 1, Figure 2) and referred to in the text, just like a table.

6.3 Calculations and the text around the data

Show one worked calculation as an example — for instance, how the rate was calculated from the time, or how the mean was found — and put the rest in an appendix. The text around the tables and figures should point out the pattern, not repeat the numbers: 'The reaction time fell steadily as the temperature rose, and the rate approximately doubled for each 10-degree rise' is useful text; 'At 20 degrees the time was 96 seconds, at 30 degrees it was 67 seconds...' is not, because the table already says that. Keep the text short, and let the tables and figures carry the detail.

- [ ] Results are presented once — in a table or a graph, not both.
- [ ] Every table and figure is numbered, titled and referred to in the text.
- [ ] Column headings and axes carry units; numbers use the right significant figures.
- [ ] Means of repeats are shown; raw data are in an appendix.
- [ ] The text points out the pattern, not every number.
- [ ] No interpretation or theory appears in this section.

Chapter 7: Discussion, Including Error Analysis

The discussion is where the report earns its marks. Results describe what happened; the discussion argues what it means. A strong discussion does four things: it states whether the results support the aim and the hypothesis; it compares the findings with the theory from the introduction; it explains the pattern — why it happened; and it analyses the errors honestly, showing how much the results can be trusted. Students often write the discussion as a second results section, repeating the numbers. Resist that. The reader has already seen the numbers; now they need your judgement.

Begin by answering the question the experiment asked. 'The results support the hypothesis: the rate of reaction increased with temperature, approximately doubling for each 10-degree rise.' Then compare the finding with the theory: does it agree with what the introduction predicted, and if so, why? The mechanism matters — for the temperature experiment, the explanation is that higher temperatures give the particles more energy, so more collisions have enough energy to react, and collisions happen more often. A discussion that only says 'the results agree with the theory' is too thin; the examiner wants to see that you understand why they agree.

7.1 Explaining unexpected results

If a result does not fit the pattern — a reading that is far from the others, a trend that flattens or reverses — do not hide it and do not pretend it did not happen. Look for a reason: was the temperature not properly controlled, was the acid concentration different, was a measurement misread? A single odd reading can often be explained by a single mistake, and the discussion should say so, and say whether the reading was included in the mean. If the explanation is not clear, say that too, and suggest what a further experiment could test. Examiners respect a student who confronts an unexpected result far more than one who smooths it away.

7.2 Error analysis

Every measurement has an uncertainty, and the discussion must say how large it is and where it comes from. There are two kinds of error. Random errors — such as the difficulty of reading a stopwatch at the exact moment a volume is reached — make repeated measurements scatter around the true value; they are reduced by taking repeats and using the mean, and they show up in the range of the repeats. Systematic errors — such as a thermometer that reads one degree too high, or a balance that was not zeroed — shift all the measurements in the same direction; they are not reduced by repeats, and they must be found and corrected or discussed.

For each main source of error, state how large it is and what effect it has on the conclusion. Use the precision of the instruments: if the thermometer reads to 0.5 degrees and the temperature was 20 degrees, the uncertainty is small; if the time was measured to the nearest second and the reaction took only 24 seconds, the percentage uncertainty is larger, and it grows as the reaction gets faster. The most important question is whether the errors could change the conclusion. If the trend is far larger than the uncertainties, the conclusion stands. If the uncertainties are as

large as the effect, the conclusion must be stated cautiously — and the discussion should say how the experiment could be improved to reduce them.

7.3 Improvements and further work

End the discussion by suggesting improvements: more repeats, a wider range of the variable, better temperature control, a more precise instrument, or a different method of measuring the same quantity. Make the suggestions specific and linked to the errors you identified — 'use a data logger to record the volume continuously, so that the timing error is removed' is a useful suggestion; 'be more careful' is not. A short suggestion for further work — what the next experiment could test — shows that you are thinking beyond the single experiment, and it is often the last thing the marker reads.

Worked Example: An error analysis that does the work (fictional experiment)

'The largest uncertainty came from the timing: the volume was read by eye and the stopwatch stopped by hand, giving an estimated uncertainty of about 1 second. At 20 degrees this is about 1 percent of the 96-second reaction time, but at 60 degrees it is about 4 percent of the 24-second time, so the results at high temperature are less precise. The temperature was controlled to within 1 degree by the water bath, which is small compared with the 10-degree steps between runs. The trend — the rate doubling for each 10-degree rise — is far larger than these uncertainties, so the conclusion is not affected by them. Using a data logger would remove the timing error and improve the precision at high temperatures.' This paragraph names the errors, sizes them, judges their effect on the conclusion, and suggests an improvement. That is the standard to aim for.

[ ] The discussion states whether the hypothesis was supported.

[ ] Findings are compared with the theory, with the mechanism explained.

[ ] Unexpected results are confronted, not hidden.

[ ] Random and systematic errors are both identified and sized.

[ ] The effect of the errors on the conclusion is judged honestly.

[ ] Specific improvements are suggested, linked to the errors found.

Chapter 8: Conclusion

The conclusion is short — usually one paragraph — and it must answer the question the experiment asked, in the light of the evidence. It states the main finding, with the key numbers, and says what it means. It does not introduce new results, new theory or new discussion; everything in the conclusion must already appear in the results and the discussion. A conclusion that repeats the aim is too weak; a conclusion that claims more than the evidence supports is too strong. The test is simple: a reader who reads only the conclusion should know what was found and whether the aim was achieved.

A good conclusion for the temperature experiment might read: 'The rate of reaction between hydrochloric acid and calcium carbonate increases with temperature, approximately doubling for each 10-degree rise between 20 and 60 degrees Celsius. The hypothesis was therefore supported, and the result is consistent with the theory that higher temperatures increase the energy and frequency of particle collisions.' Two sentences, the key finding, the key numbers, and the link to the theory. That is all a conclusion needs to do — and it is all it should try to do.

[ ] The conclusion answers the aim stated in the introduction.

[ ] The main finding is stated, with the key numbers.

[ ] Nothing new appears — no new results, theory or discussion.

[ ] The claim is no bigger than the evidence supports.

Chapter 9: References

Every idea, fact or figure in the report that came from another source must be referenced, so that the reader can check it. In an undergraduate lab report the list is usually short — a textbook, the lab manual, and perhaps one or two other sources — but the rules are the same as for any academic writing: every citation in the text must appear in the reference list, and every entry in the list must be cited in the text. A reference list with entries nobody cited, or citations with no entry, is one of the easiest marks to lose, and one of the easiest to avoid losing.

Use the referencing style your department requires — it may be a numbered style, where sources are listed in the order they are first cited, or an author-date style, where they are listed alphabetically. Whichever it is, use it consistently: the same order, the same punctuation, the same amount of detail for every entry. Record the full details of each source as you use it — author, title, edition, publisher, year, and page numbers where relevant — because finding them again a week later is far harder than writing them down at the time.

9.1 What needs a reference — and what does not

Reference the theory you used, the method you adapted from a source, and any values you quoted, such as a published constant. You do not need to reference your own measurements, your own calculations, or common knowledge in your subject — the fact that water boils at 100 degrees at standard pressure, for example. When in doubt, reference: an unnecessary reference costs nothing, while a missing one can be treated as plagiarism. Never copy sentences from a source without quotation marks and a reference; paraphrase the idea in your own words and reference the source.

Worked Example: Referencing a textbook idea

In the text: 'The rate of reaction increases with temperature because a greater proportion of collisions have energy above the activation energy.' That sentence states an idea from the theory, and it needs a reference to the textbook or source where the idea is explained. The reference list then gives the full details, so the reader can find the

source. Without the reference, the reader cannot tell whether the idea is yours, the textbook's, or a guess.

[ ] Every citation in the text appears in the reference list.

[ ] Every entry in the reference list is cited in the text.

[ ] One referencing style is used consistently throughout.

[ ] Full details of each source were recorded when it was used.

[ ] No sentence is copied from a source without quotation marks and a reference.

Chapter 10: Appendices (Raw Data, Calculations)

The appendices hold the material the reader may need to check but does not need to read to follow the report: the raw data, the full calculations, a copy of a calibration graph, or a detailed description of an instrument. They keep the main text clean and readable, while still making the evidence available. A reader who wants to check a result can turn to the appendix and find the numbers; a reader who trusts the results section can skip the appendices entirely.

Number the appendices — Appendix A, Appendix B — and give each a title. Refer to each appendix at least once in the main text, so that no appendix is orphaned: 'The raw readings are given in Appendix A, and the full calculations in Appendix B.' Do not use the appendices to hide material that belongs in the main text — a result that matters must appear in the results section, not only in an appendix — and do not include material nobody needs, such as pages of unprocessed output. An appendix should be as carefully presented as the main text, with the same headings, units and significant figures.

10.1 What belongs in an appendix

- Raw data: the individual readings from every repeat, before any averaging.
- Full calculations: every step of a calculation that is only summarised in the main text.
- Calibration data: the readings used to calibrate an instrument, with the graph if one was drawn.
- Extra tables or figures: data that support the report but would interrupt the main text.
- The risk assessment: where the department requires it, the hazards and precautions in full.

[ ] Appendices are numbered and titled.

[ ] Each appendix is referred to at least once in the main text.

[ ] Raw data and full calculations are included, clearly presented.

[ ] No important result appears only in an appendix.

[ ] No unnecessary material is included.

Chapter 11: Scientific Style, Tense and Units

Scientific writing has a style of its own: clear, precise and impersonal. The goal is not to impress the reader with long words, but to say exactly what happened and what it means, in the fewest words that do the job. Short sentences, plain words and exact numbers are the marks of good scientific writing. A sentence of twenty words that says one thing clearly is worth more than a sentence of forty words that says the same thing with decoration.

Write in the past tense when describing what you did and what you found — 'The acid was heated', 'The time was 96 seconds' — because the experiment is finished. Use the present tense for general truths and for the theory — 'The rate of reaction increases with temperature' — and for referring to the report itself: 'Table 1 shows...', 'Figure 2 illustrates...'. Mixing the tenses at random is one of the most common faults in student reports, and it is easy to fix once you know the rule: past for your work, present for general facts.

11.1 Impersonal language and precision

Scientific reports are usually written without 'I' or 'we': 'The solution was heated' rather than 'I heated the solution'. The reason is that the experiment should be repeatable by anyone, so the report describes the procedure, not the person. The passive voice is therefore common in the methods section, but it should not be used everywhere — the active voice is clearer when the actor matters, and the discussion often reads better in the active voice: 'The results support the hypothesis' is clearer than 'It is concluded by the results that the hypothesis is supported.' Use the passive where the action matters more than the actor, and the active where it does not.

Precision is the other half of style. Replace vague words with numbers: not 'the temperature was increased a lot', but 'the temperature was increased from 20 to 60 degrees Celsius'. Define a term when you first use it, and then use it consistently — do not call the same quantity 'reaction time' in one section, 'time taken' in the next, and 'duration' in the discussion. And give numbers to the right number of significant figures: a time measured to the nearest second should be written as 96 s, not 96.0 s, and a mean of three readings should not carry more figures than the readings themselves justify.

11.2 Units and symbols

Use the SI system of units throughout, and write them correctly: a space between the number and the unit ('50 cm3', '20 degrees Celsius'), the unit in its standard symbol, and the same unit for the same quantity throughout the report. Do not mix units — if the temperature is in degrees Celsius in the methods, it must be in degrees Celsius in the results and the discussion. Give the unit in every column heading and on every graph axis, so that the reader never has to guess. A number without a unit is meaningless in science, and a marker will treat it as an error.

Quantity	Unit and symbol	Common mistake

Volume	cubic centimetres (cm3)	Writing 'ml' in one place and 'cm3' in another
Temperature	degrees Celsius (C)	Forgetting the unit on a graph axis
Time	seconds (s)	Giving a time to more figures than the stopwatch allows
Mass	grams (g)	Mixing grams and kilograms in one table
Concentration	mol/dm3	Stating a concentration without its unit

- [ ] Past tense for what was done and found; present tense for general facts.
- [ ] Impersonal language in the methods; clear active voice where it helps.
- [ ] Vague words replaced by numbers throughout.
- [ ] Key terms defined once and used consistently.
- [ ] SI units used correctly, with a space between number and unit.
- [ ] Numbers given to the right number of significant figures.

Chapter 12: Worked Example — a Complete Fictional Experiment Report

This chapter presents a complete short lab report for a fictional experiment, so that you can see the whole IMRaD structure working together. The experiment, the data and the results are invented for teaching purposes: the numbers are plausible, but no real experiment was performed, and the report should not be quoted as evidence about the real world. Read it as a model of structure and style, not as a source of facts. Notice how each section does only its own job, how the tables and the text work together, and how the discussion handles errors honestly.

12.1 Title and abstract

Worked Example: Title

The Effect of Temperature on the Rate of Reaction between Hydrochloric Acid and Calcium Carbonate

Abstract: This experiment investigated how temperature affects the rate of reaction between hydrochloric acid and calcium carbonate. Five temperatures between 20 and 60 degrees Celsius were tested, and the time taken to collect 50 cm3 of carbon

dioxide was measured at each temperature, with three repeats. The mean reaction time fell from 96 seconds at 20 degrees to 24 seconds at 60 degrees, and the rate approximately doubled for each 10-degree rise. The results support the hypothesis that higher temperatures increase the rate of reaction, consistent with the theory that more particles have energy above the activation energy at higher temperatures.

12.2 Introduction

The rate of a chemical reaction depends on several factors, including the concentration of the reactants, the surface area of any solid, and the temperature. The theory of reaction rates states that particles must collide with enough energy to react, and that the proportion of collisions with enough energy rises as the temperature rises. This experiment tested that prediction for one reaction: the reaction between hydrochloric acid and calcium carbonate, which produces carbon dioxide gas. The aim was to investigate how temperature affects the rate of this reaction. It was predicted that the rate would increase with temperature.

12.3 Methods

50 cm3 of 1.0 mol/dm3 hydrochloric acid was placed in a conical flask in a water bath at the required temperature. 5.0 g of calcium carbonate chips were added, the flask was stoppered, and the carbon dioxide produced was collected in a gas syringe. The volume was recorded every 10 seconds until 50 cm3 had been collected, and the time was noted. The procedure was repeated at 30, 40, 50 and 60 degrees Celsius, with three repeats at each temperature. The acid concentration, the mass of calcium carbonate and the volume of gas collected were kept constant. Eye protection was worn throughout, and the acid was handled with care because it is corrosive.

12.4 Results

Table 1 (in Chapter 6 of this guide) shows the results: the mean time to collect 50 cm3 of carbon dioxide fell from 96 seconds at 20 degrees Celsius to 24 seconds at 60 degrees. A graph of the rate against temperature would show the rate rising steadily, approximately doubling for each 10-degree rise. The three repeats at each temperature agreed closely, with a range of no more than 3 seconds at any temperature.

12.5 Discussion and conclusion

The results support the hypothesis: the rate of reaction increased with temperature, approximately doubling for each 10-degree rise, as predicted by the theory that a greater proportion of collisions have enough energy to react at higher temperatures. The largest uncertainty came from the timing, estimated at about 1 second; at 60 degrees this is about 4 percent of the 24-second reaction time, but the trend is far larger than the uncertainty, so the conclusion is not affected. Using a data logger to record the volume continuously would remove the timing error. In conclusion, the rate of reaction between hydrochloric acid and calcium carbonate increases with temperature, approximately doubling for each 10-degree rise between 20 and 60 degrees Celsius.

[ ] The title names the variable and the reaction studied.

[ ] The abstract gives aim, method, results and conclusion in five sentences.

[ ] The introduction states the theory, the aim and the prediction.

[ ] The methods could be repeated by another student.

[ ] The results present the data without interpreting them.

[ ] The discussion interprets the results, analyses the errors, and suggests an improvement.

Chapter 13: Common Mistakes and a Checklist

The mistakes below appear in lab report after lab report, in every science subject. They are not mistakes of ability; they are mistakes of habit, and every one of them is avoidable. Read this chapter before you submit, when there is still time to fix what it finds.

13.1 The ten most common mistakes

• Writing the report as a diary — 'First I did this, then I did that' — instead of organising it by section.

• Copying the lab manual into the methods section instead of describing what was actually done.

• A title that says nothing, such as 'Chemistry Experiment 3'.

• An abstract that describes the experiment but never gives the results.

• Presenting the same data in both a table and a graph.

• Repeating every number from a table in the text, instead of pointing out the pattern.

• Mixing results and interpretation, so the reader cannot tell evidence from opinion.

• Hiding an unexpected result instead of explaining it.

• No error analysis — or an error analysis that lists errors without sizing them.

• Numbers without units, or with more significant figures than the instrument allows.

• References missing, incomplete, or never cited in the text.

• Leaving the writing to the night before the deadline, when there is no time to check it.

13.2 The final checklist

Work through this checklist with the printed report in front of you. Tick each item only when you have checked it on the page, not from memory.

[ ] Title, abstract, introduction, methods, results, discussion and conclusion are all present.

[ ] The aim is stated in the introduction and answered in the conclusion.

[ ] The methods would allow another student to repeat the experiment.

[ ] Every table and figure is numbered, titled, referred to in the text, and labelled

with units.

[ ] No data appear in both a table and a figure.

[ ] The discussion interprets the results and analyses the errors honestly.

[ ] Every number has a unit and the right number of significant figures.

[ ] Every citation appears in the reference list, and every reference is cited.

[ ] Appendices are numbered, titled and referred to in the text.

[ ] The report has been read through once, aloud, for sense and for errors.

[ ] The file is saved in the required format, with a backup copy kept separately.

Chapter 14: Conclusion

A good lab report is not a long report; it is a clear one. The structure in this guide — title, abstract, introduction, methods, results, discussion, conclusion, references, appendices — exists so that the reader can find what they need in seconds and judge the work fairly. Learn the structure once, and every lab report you write will be easier than the last, because you will never have to wonder what goes where.

The habits that make a good report are the habits of good science: record measurements honestly as you take them, describe the method so that it can be repeated, present the results without hiding anything, interpret them carefully, and analyse the errors honestly. A report written that way is useful even when the experiment did not work as expected — because an honest report of an unexpected result is worth more than a tidy report of invented data.

Write the methods while the experiment is fresh, the results while the analysis is in front of you, and the introduction and abstract last, when you know what the report claims. Then read the whole report once, aloud, check it against the checklist in Chapter 13, and submit it with the confidence of someone who knows exactly what it says and why. Good luck — and keep good notebooks.

Suggested Reading

The following are well-known general works on scientific writing and laboratory practice that students commonly consult. Check your university library for the latest editions, and follow your department's guidance on referencing style.

- General guides to scientific writing: widely used books on writing lab reports and research papers explain the IMRaD structure, style, tables, figures and referencing, with examples from the sciences.
- Laboratory practice guides: standard guides to working in a teaching laboratory cover safety, measurement, recording data and handling uncertainties.
- Statistics and data handling guides: general textbooks on data analysis for science students explain means, errors, significant figures and the presentation of results.
- Style manuals: the official manuals of the major referencing styles (APA, Harvard, Chicago, and the numbered styles used in the sciences) are the authoritative source for citation and formatting rules — consult the edition your department requires.

- Your department's lab report guide and template: always the first and final authority on structure, word count, formatting and submission requirements for your course.
- Your teacher's or demonstrator's guidance for the specific experiment: the most relevant source of all, because it is written for the apparatus and the marking scheme you are actually using.